

## KIRKJUBAEJARKLAUSTUR, Iceland

WMO: 040640

Lat: 63.793N

Lon: 18.012W

Elev: 22

StdP: 101.06

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | -8.4       | -6.9 | -15.8                         | 0.9 | -7.5 | -13.9 | 1.1 | -6.1 | 15.2                  | -2.3 | 12.8 | 1.4  | 3.9                   | 340  | 0.725 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 5.3                    | 18.0            | 13.6 | 16.6 | 12.7 | 15.3 | 12.0 | 14.3                | 17.0 | 13.3 | 15.4 | 12.6 | 14.4 | 4.4                  | 200  |

| Dehumidification DP/MCDB and HR |     |      |      |     |      |      |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-----|------|------|-----|------|------|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |     |      | 1%   |     |      | 2%   |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR  | MCDB | DP   | HR  | MCDB | DP   | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b) | (c)  | (d)  | (e) | (f)  | (g)  | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 13.0                            | 9.3 | 15.0 | 12.2 | 8.9 | 14.0 | 11.6 | 8.5 | 13.2 | 40.1          | 17.3 | 37.4 | 15.6 | 35.6 | 14.5 | 19.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 14.0          | 11.9 | 9.3 | DB                         | -10.9 | 21.5               | 1.7 | 1.7                                                | -12.1 | 22.7       | -13.0 | 23.7       | -14.0 | 24.7       | -15.2 | 26.0 | (4) |
| (5)               |      |     | WB                         | -12.1 | 16.8               | 1.6 | 1.4                                                | -13.2 | 17.8       | -14.1 | 18.6       | -15.0 | 19.4       | -16.1 | 20.4 | (5) |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 5.3   | 0.7   | 0.2   | 1.1   | 4.1   | 7.1   | 10.1  | 11.9  | 11.5  | 8.6   | 4.9   | 2.0   | 0.5  | (6)  |
| (7)  |                                                                                  | DBStd       | 5.13  | 3.26  | 3.41  | 3.37  | 3.04  | 2.48  | 2.19  | 1.62  | 1.70  | 2.40  | 3.00  | 3.25  | 3.35 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 1888  | 288   | 273   | 276   | 178   | 95    | 24    | 2     | 4     | 53    | 159   | 241   | 294  | (8)  |
| (9)  |                                                                                  | HDD18.3     | 4772  | 547   | 507   | 534   | 428   | 349   | 246   | 198   | 211   | 292   | 417   | 491   | 552  | (9)  |
| (10) |                                                                                  | CDD10.0     | 157   | 0     | 0     | 0     | 0     | 4     | 28    | 62    | 52    | 11    | 0     | 0     | 0    | (10) |
| (11) |                                                                                  | CDD18.3     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 1     | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0     | 0     | 0     | 0     | 0    | (12) |
| (13) |                                                                                  | CDH26.7     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 3.7   | 3.8   | 3.6   | 3.9   | 4.2   | 3.6   | 3.5   | 3.2   | 3.3   | 3.8   | 4.0   | 3.8   | 3.7  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 1993  | 185   | 182   | 156   | 142   | 115   | 111   | 132   | 188   | 209   | 198   | 171   | 218  | (15) |
| (16) |                                                                                  | PrecMax     | 2658  | 384   | 432   | 305   | 312   | 211   | 196   | 285   | 361   | 373   | 451   | 330   | 324  | (16) |
| (17) |                                                                                  | PrecMin     | 1461  | 53    | 49    | 57    | 43    | 20    | 24    | 42    | 89    | 118   | 89    | 34    | 101  | (17) |
| (18) |                                                                                  | PrecStd     | 247   | 83    | 92    | 66    | 71    | 38    | 55    | 62    | 70    | 67    | 88    | 69    | 66   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 8.5   | 8.8   | 9.0   | 11.9  | 16.7  | 19.9  | 20.1  | 15.4  | 11.4  | 9.4   | 8.7   | (19) |      |
| (20) |                                                                                  |             | MCWB  | 7.8   | 8.5   | 7.5   | 7.9   | 11.4  | 14.4  | 14.7  | 15.9  | 11.9  | 9.6   | 8.5   | 8.3  | (20) |
| (21) |                                                                                  | 2%          | DB    | 6.9   | 7.2   | 7.8   | 10.0  | 13.9  | 17.2  | 18.1  | 17.3  | 13.6  | 10.6  | 8.4   | 7.5  | (21) |
| (22) |                                                                                  |             | MCWB  | 6.1   | 6.7   | 6.6   | 7.1   | 9.8   | 12.4  | 13.6  | 13.7  | 11.3  | 9.6   | 7.7   | 7.0  | (22) |
| (23) |                                                                                  | 5%          | DB    | 5.8   | 6.1   | 6.9   | 9.0   | 12.1  | 15.5  | 16.7  | 15.8  | 12.6  | 10.0  | 7.6   | 6.3  | (23) |
| (24) |                                                                                  |             | MCWB  | 5.0   | 5.4   | 5.9   | 7.0   | 8.8   | 11.5  | 12.9  | 12.6  | 10.8  | 9.2   | 6.9   | 5.6  | (24) |
| (25) |                                                                                  | 10%         | DB    | 4.8   | 5.0   | 5.7   | 8.2   | 10.8  | 14.0  | 15.3  | 14.6  | 11.8  | 9.1   | 6.7   | 5.2  | (25) |
| (26) |                                                                                  |             | MCWB  | 3.8   | 4.3   | 4.5   | 6.6   | 8.1   | 10.7  | 12.2  | 11.9  | 10.3  | 8.2   | 5.8   | 4.2  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 7.9   | 8.3   | 8.1   | 9.1   | 12.2  | 15.0  | 15.6  | 16.2  | 13.0  | 10.7  | 9.1   | 8.5  | (27) |
| (28) |                                                                                  |             | MCDWB | 8.1   | 8.7   | 8.7   | 10.3  | 16.0  | 18.8  | 18.8  | 19.0  | 13.9  | 10.8  | 9.2   | 8.6  | (28) |
| (29) |                                                                                  | 2%          | WB    | 6.4   | 6.8   | 7.1   | 8.3   | 10.5  | 13.0  | 14.2  | 14.4  | 12.1  | 10.0  | 8.0   | 7.1  | (29) |
| (30) |                                                                                  |             | MCDWB | 6.7   | 7.1   | 7.6   | 9.1   | 12.9  | 16.1  | 17.0  | 16.7  | 13.0  | 10.3  | 8.2   | 7.4  | (30) |
| (31) |                                                                                  | 5%          | WB    | 5.2   | 5.6   | 6.1   | 7.6   | 9.5   | 11.9  | 13.3  | 13.2  | 11.4  | 9.4   | 7.1   | 5.9  | (31) |
| (32) |                                                                                  |             | MCDWB | 5.7   | 6.0   | 6.8   | 8.5   | 11.3  | 14.7  | 15.7  | 14.9  | 12.1  | 9.8   | 7.5   | 6.2  | (32) |
| (33) |                                                                                  | 10%         | WB    | 4.1   | 4.4   | 4.8   | 6.9   | 8.7   | 11.1  | 12.6  | 12.5  | 10.7  | 8.5   | 5.9   | 4.5  | (33) |
| (34) |                                                                                  |             | MCDWB | 4.7   | 4.9   | 5.6   | 7.9   | 10.2  | 13.3  | 14.6  | 13.9  | 11.4  | 9.0   | 6.5   | 5.0  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 3.8   | 4.3   | 4.7   | 5.4   | 5.7   | 5.7   | 5.3   | 5.4   | 4.7   | 4.4   | 3.8   | 3.9  | (35) |
| (36) |                                                                                  |             | MCDBR | 4.5   | 4.9   | 5.0   | 5.6   | 7.7   | 8.8   | 8.6   | 7.7   | 5.6   | 4.5   | 4.1   | 4.7  | (36) |
| (37) |                                                                                  |             | MCWBR | 4.5   | 4.9   | 4.4   | 4.1   | 4.9   | 5.5   | 5.4   | 5.2   | 4.2   | 4.1   | 4.2   | 4.6  | (37) |
| (38) |                                                                                  | 5% WB       | MCDWB | 4.5   | 4.8   | 4.6   | 4.6   | 6.4   | 7.6   | 7.0   | 6.5   | 4.3   | 3.9   | 3.9   | 4.7  | (38) |
| (39) |                                                                                  |             | MCWBR | 4.7   | 4.9   | 4.4   | 3.8   | 4.5   | 5.1   | 4.8   | 4.9   | 3.8   | 4.0   | 4.2   | 4.8  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.268 | 0.315 | 0.340 | 0.366 | 0.369 | 0.382 | 0.365 | 0.357 | 0.355 | 0.317 | 0.393 | 0.396 | (40) |      |
| (41) |                                                                                  | taud        | 1.938 | 2.114 | 2.158 | 2.167 | 2.255 | 2.276 | 2.385 | 2.400 | 2.304 | 2.200 | 2.654 | 2.486 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 272   | 576   | 730   | 800   | 838   | 833   | 841   | 813   | 719   | 567   | 313   | 129   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 32    | 66    | 94    | 116   | 116   | 117   | 102   | 92    | 82    | 59    | 28    | 18    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.14  | 0.65  | 1.85  | 3.70  | 5.24  | 5.55  | 4.88  | 3.88  | 2.24  | 0.96  | 0.24  | 0.06  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.01  | 0.09  | 0.16  | 0.35  | 0.43  | 0.46  | 0.33  | 0.23  | 0.17  | 0.10  | 0.03  | 0.00  | (45) |      |

## Historical Trends

|                             |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |  |  |
|-----------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|--|--|
|                             |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |  |  |
|                             |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |  |  |
| (46) Station Only           |  | N/A   | N/A     | +1.88  | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |  |  |
| (47) Regional (6 neighbors) |  | +0.60 | +1.11   | +1.30  | +0.63   | N/A   | N/A   | -169        | -225    | +45     | N/A     |  |  |

Nomenclature: See separate page